Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

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Abstract

Classical regime detection methods operate in flat Euclidean feature spaces, relying on distributional assumptions that are violated during the very transitions they seek to detect. We construct a learned Riemannian manifold for financial markets by lifting time series into a projective Hilbert space $\mathbb{CP}^{n-1}$ via a parameterized error Hamiltonian and extracting the induced geometric structure: Fubini–Study metric tensor, Berry curvature, spectral gap, and Fisher information matrix. This geometry induces a family of scalar observables—Berry phase rate, Fisher information pseudo-determinant, rolling curvature integrals, multi-lag fidelity, and metric condition number—that summarize the local and global structure of the market manifold at each point in time.

In a head-to-head comparison of 17 methods across 12 historical crises (2007–2023), the three best geometric observables—Berry Phase Rate ($d = 0.93$), QFI Determinant ($d = 0.93$), and Multi-Lag Fidelity ($d = 0.84$)—are *competitive with but do not individually exceed* the supervised Random Forest baseline ($d = 1.13$) after Holm–Bonferroni correction. The geometric methods are, however, fully *unsupervised* and require no labeled crisis data. They also substantially outperform two standard unsupervised baselines: BOCPD ($d = 0.59$) and Isolation Forest ($d = 0.46$).

Our central finding is *complementarity*: the geometric and supervised signals detect structurally different crisis types. **(i)** On crises where RF underperforms (Volmageddon, SVB, Fed Selloff), geometric methods achieve $d > 1.0$ while RF scores $d < 0.2$. **(ii)** A simple hybrid ensemble nearly doubles out-of-sample effect size relative to RF alone ($d = 0.60$ vs. $d = 0.31$). **(iii)** Geometric observables generalize across five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Multi-Lag Fidelity $d = 1.44$ vs. RF $d = 0.88$ (Friedman $p < 0.0001$). **(iv)** Optimized multi-observable fusion achieves median $d = 1.67$, though nested leave-one-crisis-out validation reveals an overfitting gap (held-out $d = 1.09$), underscoring the importance of unbiased evaluation.

These findings establish the learned market manifold as a principled geometric framework that complements supervised approaches, with greatest value precisely where pattern-matching methods lack historical precedent.

Keywords: regime detection, spectral geometry, Riemannian metric, Berry curvature, Chern number, Fisher information, financial crises, metric learning.

1 Introduction

Detecting regime transitions is a central problem in quantitative finance. Market dynamics alternate between qualitatively different states—low-volatility trends, high-volatility drawdowns, liquidity crises—and the ability to identify transitions in real time has implications for risk management, portfolio construction, and hedging.

Classical approaches include Hidden Markov Models (HMMs) [1], cumulative-sum (CUSUM) statistics [2], rolling volatility breakpoints, and supervised classifiers such as Random Forests. While effective in specific settings, these methods share a common limitation: they operate entirely within a Euclidean feature space and rely on distributional assumptions (Gaussian mixtures, stationarity within regimes) that are frequently violated during the very transitions they seek to detect.

Our central thesis is that financial regimes leave distinct geometric signatures on a learned data manifold, and that regime transitions manifest as geometric singularities—curvature spikes, spectral gap closures, and rapid changes in curvature integrals—of this manifold.

Why lift data into a curved space, rather than applying a simpler nonlinear method such as a kernel machine or neural network? The projective Hilbert space embedding provides several structural advantages. *First*, the Berry curvature F_{ab} is gauge-invariant—unchanged under unitary rotations $|\psi\rangle \mapsto e^{i\theta}|\psi\rangle$—whereas kernel-based distances depend on the choice of coordinate representation. *Second*, the spectral gap $\Delta(x)$ between the ground and first excited state closes *continuously* before a geometric transition, providing a principled leading indicator with no ad hoc threshold selection. *Third*, the non-commutativity of the Hermitian operators $\{O_j\}$ encodes cross-feature interactions (e.g., momentum–volatility coupling) in the off-diagonal structure of $H(x)$, capturing higher-order dependencies that PCA-based methods, which diagonalize by construction, necessarily discard. *Fourth*, for closed parameter surfaces the integral of Berry curvature yields a Chern number $c_1 \in \mathbb{Z}$, a topological invariant; however, as we discuss in Section 8, our rolling-window estimates are computed over open domains and are therefore non-integer, limiting the strength of topological protection arguments in practice.

In this paper we construct a learned Riemannian manifold for financial markets and show that regime detection emerges as a natural consequence of its geometry. Inspired by the Quantum Cognition Machine Learning (QCML) framework [3], we *lift* the feature space $\mathbb{R}^d$ into a projective Hilbert space $\mathbb{CP}^{n-1}$ via a family of Hermitian operators. We emphasize that this construction uses the *mathematical machinery* of Hilbert spaces and spectral theory—not quantum mechanics proper. There is no quantum superposition, entanglement, or quantum computing involved; the connection to physics is purely formal. The ground state $|\psi(x)\rangle$ of an error Hamiltonian $H(x)$ defines a smooth map whose differential geometry provides a rich set of observables:

1. The *Fubini–Study metric tensor* g_{ab} (the real part of the Fubini–Study metric pulled back to $\mathbb{R}^d$) endows the data manifold with a natural Riemannian structure, replacing the ad hoc Euclidean metric.

2. The *Berry curvature* F_{ab} (the antisymmetric, imaginary part) [13] encodes information about the non-trivial geometry of the ground-state bundle over parameter space. For closed surfaces, its integral yields integer-valued Chern numbers; for rolling windows over open domains, it provides a continuous curvature signal.

3. The *spectral gap*, *fidelity*, and *Fisher information* observables probe different aspects of the manifold's local and global structure, providing complementary detection channels.

Contributions.

1. **The geometry.** We construct the full Riemannian geometry of a financial data manifold via a parameterized error Hamiltonian, including the Fubini–Study metric, Berry curvature, Christoffel symbols, Ricci scalar, spectral gap, and Fubini–Study distances (Section 3).

2. **Geometric observables.** The geometry induces a family of scalar observables—Berry phase rate, Fisher information pseudo-determinant, curvature integrals, multi-lag fidelity—that summarize local and global manifold structure at each point in time (Section 4).

3. **Empirical validation.** In a comparison of 16 methods across 12 historical crises, the best geometric observables are competitive with—though do not individually exceed—the supervised Random Forest baseline. Our key finding is *complementarity*: geometric and supervised methods detect structurally different crisis types, and a simple hybrid ensemble nearly doubles out-of-sample performance relative to either approach alone (Section 6).

2 Background and Related Work

2.1 Classical Regime Detection

The regime-switching framework originates with Hamilton's Markov-switching autoregression [1], in which a latent discrete state governs the parameters of an observed process. The two-state Gaussian HMM remains a workhorse, estimating $P(\text{high-vol state} \,|\, r_t)$ via the Baum–Welch algorithm. CUSUM statistics [2] provide a nonparametric alternative, accumulating deviations $S_t = \max(0, S_{t-1} + |r_t| - k)$ and flagging $S_t > h$ as a changepoint. Rolling volatility z-scores are perhaps the simplest approach: one computes $z_t = (\sigma_t^{\text{short}} - \mu_t^{\text{long}})/\sigma_t^{\text{long}}$ and applies a threshold. Supervised classifiers (Random Forest, gradient-boosted trees) can achieve high in-sample accuracy but require labeled data and are prone to overfitting on rare events [4].

Bayesian online changepoint detection. Adams and MacKay [14] introduced Bayesian Online Changepoint Detection (BOCPD), which maintains a posterior distribution over the *run length*—the time since the last changepoint—and updates it recursively as new observations arrive. Extensions to multivariate settings and non-exponential-family likelihoods have been explored by Knoblauch and Damoulas [17]. While BOCPD provides principled uncertainty quantification over changepoint timing, it remains a distributional method operating in flat Euclidean space.

All of these methods treat features as elements of a flat Euclidean space. Our approach differs by constructing a learned Riemannian manifold whose curvature and topology encode structural information about the market state—and whose geometric observables, as we show, change sharply at regime boundaries.

2.2 Spectral Metric Learning

The QCML framework [3] embeds classical data into the mathematical structures of spectral theory. Each data point $x \in \mathbb{R}^d$ is associated with a Hermitian matrix $H(x)$ whose eigenvector $|\psi(x)\rangle$ serves as a nonlinear feature map $\mathbb{R}^d \to \mathbb{CP}^{n-1}$. This is conceptually analogous to kernel methods—the Fubini–Study metric induces a positive-definite kernel $K(x, y) = |\langle\psi(x)|\psi(y)\rangle|^2$—but additionally provides geometric structure (metric tensor, curvature, topological invariants) absent from standard kernel machines. We stress that this construction is purely mathematical: while we borrow notation and terminology from quantum mechanics for historical reasons ($|\psi\rangle$, "Hamiltonian", "ground state"), the framework involves no quantum physics.

2.3 Topological Data Analysis in Finance

Topological data analysis (TDA) uses persistent homology to detect early warning signals of market crashes. Gidea and Katz [18] computed Betti numbers from sliding-window point clouds of daily returns, showing that persistent homology norms spike $\sim$5 trading days before crashes. Gidea [19] extended this to financial correlation networks. Our approach differs in that we extract topology from a *learned* spectral structure (Berry curvature and Chern numbers of a parameterized eigenproblem) rather than from point-cloud pairwise distances, providing a richer geometric structure (metric tensor, curvature, spectral gap) beyond topology alone.

2.4 Deep Learning Approaches to Regime Detection

Deep learning methods—including LSTMs [20], Temporal Convolutional Networks [21], Transformer-based architectures, and autoencoder-based anomaly detection [22]—have been applied to regime detection. While attention mechanisms can capture long-range dependencies, these approaches remain fundamentally supervised, requiring labeled data and tending to overfit on the small number of historical crises available. Our geometric observables are unsupervised, detecting regime transitions as intrinsic geometric singularities without labeled training data.

2.5 Geometric Methods in Finance

Information geometry [7] studies the differential geometry of statistical models. The Fisher information metric on the space of probability distributions has been applied to financial time series analysis [8], where geodesic distances between estimated distributions measure regime dissimilarity. Our metric g_{ab} is the natural generalization of the Fisher metric to the projective Hilbert space setting: when the parametric family $|\psi(x)\rangle$ is interpreted as a statistical model (via Born's rule), g_{ab} reduces to $\frac{1}{4}$ times the Fisher information matrix [6].

Rough paths and path signatures. The theory of rough paths [23] provides a coordinate-free description of irregular paths, and the *path signature*—the iterated integrals of a path—serves as a universal nonlinear feature map. Chevyrev and Kormilitzin [24] applied signatures to sequential data classification, and Lyons et al. [25] demonstrated their effectiveness for financial time series analysis. The path signature captures order-dependent information (lead-lag relationships) that Euclidean features miss. Our approach is complementary: while signatures characterize the path in a feature space, our geometry characterizes the *manifold structure* induced by projecting into Hilbert space.

2.6 Spectral Methods in Finance

Several authors have applied mathematical structures from quantum theory to finance, including path-integral option pricing [26] and quantum probability models for decision-making [27]. Our work is distinct from these in that we make no claim about quantum mechanics governing markets. We use the *mathematical machinery* of Hilbert spaces, Hermitian operators, and Berry phases purely as a structured nonlinear embedding that provides geometric tools (metric tensor, curvature, spectral gap) unavailable in flat feature spaces. The term "quantum" in our framework name (QCML) is a historical artifact reflecting the origin of these mathematical tools; the method itself is entirely classical.

3 Learned Market Geometry

The central mathematical object of this paper is a Riemannian manifold learned from financial data. We construct it by lifting the feature space into a projective Hilbert space via a parameterized error Hamiltonian, and then extracting the induced geometric structure—Fubini–Study metric, Berry curvature, spectral gap, and higher invariants—as properties of the market state space. All constructions are implemented in the `QCMLGeometry` class; we include code-level references for reproducibility.

3.1 Error Hamiltonian and Quasi-Coherent States

Let $\{A_k\}_{k=1}^d$ be a family of $n \times n$ Hermitian operators (observables) learned from data. Given a data point $x \in \mathbb{R}^d$, we define the *error Hamiltonian*:

$$H(x) = \frac{1}{2} \sum_{k=1}^{d} (A_k - x_k\, I)^2 \,, \tag{1}$$

where I is the $n \times n$ identity matrix. The Hamiltonian $H(x)$ is Hermitian and positive semi-definite by construction. Intuitively, $H(x)$ measures the collective "error" between the observables $\{A_k\}$ and the target eigenvalues $\{x_k\}$: a data point x that is well-represented by the operators yields a small ground state energy.

Definition 1 (Quasi-coherent state)**.** *The* quasi-coherent state *at x is the ground state of $H(x)$:*

$$|\psi(x)\rangle = \underset{|\phi\rangle,\, \|\phi\|=1}{\arg\min}\, \langle \phi | H(x) | \phi \rangle \,. \tag{2}$$

The map $x \mapsto |\psi(x)\rangle$ is a smooth embedding $\mathbb{R}^d \hookrightarrow \mathbb{CP}^{n-1}$ wherever the ground state is non-degenerate (i.e., the spectral gap $\Delta(x) > 0$). This map lifts each market observation into the projective Hilbert space, endowing the data manifold with Riemannian and topological geometry.

Theorem 2 (Smoothness of the QCML embedding)**.** *Let $\{A_k\}_{k=1}^d$ be Hermitian operators and let $H(x)$ be the error Hamiltonian* (1)*. If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ for all x in an open set $U \subset \mathbb{R}^d$, then:*

1. *The ground state map $x \mapsto |\psi(x)\rangle \in \mathbb{CP}^{n-1}$ is C^∞ on U.*
2. *The metric tensor $g_{ab}(x)$ and Berry curvature $F_{ab}(x)$ are C^∞ on U.*
3. *All geometric observables derived from g_{ab} and F_{ab} (Christoffel symbols, Ricci scalar, Chern numbers) are C^∞ on U.*

The proof follows from the implicit function theorem applied to the eigenvalue equation $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$: the non-degeneracy condition $\Delta > 0$ ensures that the ground state depends smoothly on parameters. See Appendix A for details.

Operator learning. The operators $\{A_k\}$ are constructed from the data covariance structure. In the *PCA-inspired* method, we compute the principal components of the data matrix X, then scale Pauli-basis operators by the corresponding eigenvalues: $A_k = \sqrt{\lambda_k}\, \sigma_k$, where σ_k are tensor products of Pauli matrices (for $n = 4$, the two-qubit basis $\{I \otimes I, I \otimes X, \ldots, Z \otimes Z\}$). Alternative methods include random Hermitian matrices and pure Pauli bases.

3.2 Fubini–Study Metric Tensor

The pullback of the Fubini–Study metric [12] to $\mathbb{R}^d$ defines the *metric tensor*:

$$g_{ab}(x) = \mathrm{Re}\,\langle \partial_a\psi|\partial_b\psi\rangle - \langle \partial_a\psi|\psi\rangle\langle\psi|\partial_b\psi\rangle\,, \tag{3}$$

where $\partial_a \equiv \partial/\partial x^a$. This is a real, symmetric, positive semi-definite tensor that generalizes the Fisher information metric to the Hilbert space setting. It measures the sensitivity of the ground state to infinitesimal changes in the data coordinates: large g_{ab} indicates that nearby data points map to distant states in $\mathbb{CP}^{n-1}$.

Partial derivatives are computed numerically via central finite differences:

$$|\partial_a\psi(x)\rangle \approx \frac{|\psi(x+\epsilon\, e_a)\rangle - |\psi(x-\epsilon\, e_a)\rangle}{2\epsilon}\,, \tag{4}$$

with $\epsilon = 10^{-5}$.

3.3 Berry Curvature

The antisymmetric part of the geometric tensor defines the *Berry curvature* [9, 11]:

$$F_{ab}(x) = -2\,\mathrm{Im}\,\langle \partial_a\psi|\partial_b\psi\rangle\,. \tag{5}$$

This is a closed 2-form on $\mathbb{R}^d$ that encodes topological information. While the metric tensor g_{ab} is sensitive to parameterization, the integral of F_{ab} over closed surfaces yields topological invariants (Chern numbers) that are integers [10] and therefore robust to smooth deformations.

For two-dimensional subspaces, we also implement a *plaquette* (Wilson loop) method that is more numerically stable for computing Chern numbers. The Berry phase around a plaquette with corners $x_{00}, x_{10}, x_{11}, x_{01}$ is:

$$\gamma = \mathrm{Im}\,\log\bigl(\langle\psi_{00}|\psi_{10}\rangle\langle\psi_{10}|\psi_{11}\rangle\langle\psi_{11}|\psi_{01}\rangle\langle\psi_{01}|\psi_{00}\rangle\bigr)\,, \tag{6}$$

where the argument of the logarithm is the Wilson loop operator. The Berry curvature is then $F_{ab} \approx \gamma/\epsilon^2$.

Theorem 3 (Chern number quantization)**.** *Let $S \subset \mathbb{R}^d$ be a closed, oriented 2-surface in parameter space over which the ground state is non-degenerate ($\Delta > 0$ on S). Then the first Chern number*

$$C_1 = \frac{1}{2\pi}\oint_S F_{ab}\,dx^a \wedge dx^b \;\in\; \mathbb{Z} \tag{7}$$

is an integer. In particular, C_1 is invariant under smooth deformations of the Hamiltonian parameters that preserve $\Delta > 0$.

This is a consequence of the Chern–Weil theorem applied to the $U(1)$ line bundle over S defined by the ground state $|\psi(x)\rangle$; see Appendix A for the proof.

Important caveat: open windows. The quantization theorem requires integration over a *closed* surface S. In our empirical application, we compute curvature integrals over *rolling windows*—open subsets of parameter space—yielding *non-integer* values (e.g., 0.82, 1.15). The topological protection argument therefore does not strictly apply to our rolling estimates. Nevertheless, the curvature integrals remain informative: their magnitude reflects the degree of geometric change within each window, and sharp transitions between windows are empirically associated with regime boundaries. We use the term "Chern number" loosely to refer to these curvature integrals, acknowledging that strict topological quantization holds only in the closed-surface limit.

3.4 Spectral Gap

The *spectral gap* at x is the difference between the two lowest eigenvalues of the error Hamiltonian:

$$\Delta(x) = E_1(x) - E_0(x)\,. \tag{8}$$

A large gap indicates a well-defined, non-degenerate ground state and hence stable geometry. A closing gap signals the approach to a *critical point*—in our financial analogy, a regime transition [5]. The spectral gap also controls the accuracy of the adiabatic approximation: when $\Delta \to 0$, the system is susceptible to non-adiabatic transitions between eigenstates.

Theorem 4 (Curvature–gap bound)**.** *For any pair of indices (a, b) with $a \neq b$, the Berry curvature is bounded by the spectral gap:*

$$|F_{ab}(x)| \;\leq\; \frac{K}{\Delta(x)^2}\,, \tag{9}$$

where $K = 2\sum_{n\neq 0}\big|\langle n|\partial_a H|0\rangle\big|\,\big|\langle n|\partial_b H|0\rangle\big|$ depends on the Hamiltonian derivatives but not on the gap.

This bound follows from first-order perturbation theory for the Berry connection (see Appendix A). It formalizes the mechanism underlying curvature-based regime detection: as the spectral gap $\Delta \to 0$ at a phase boundary, the Berry curvature can diverge as Δ^{-2}, producing the sharp spikes that our detectors identify. The converse also holds qualitatively: in stable regimes where Δ is large, curvature fluctuations are suppressed.

3.5 Fisher Information Susceptibility

The *Fisher information susceptibility* is defined as the trace of the metric tensor:

$$\chi(x) = \mathrm{tr}\big(g_{ab}(x)\big) = \sum_a g_{aa}(x)\,. \tag{10}$$

This scalar quantity equals $\frac{1}{4}$ times the total Fisher information and measures the aggregate sensitivity of the ground state to parameter changes. Near regime transitions, the market state becomes maximally sensitive to perturbations, causing χ to spike—analogous to divergent susceptibility at critical points in statistical mechanics [6].

We further define the *QFI pseudo-determinant* as

$$\overset{+}{\det}(g) = \prod_{\lambda_i > \epsilon} \lambda_i\,, \tag{11}$$

where $\{\lambda_i\}$ are the eigenvalues of g_{ab} and $\epsilon = 10^{-10}$ filters near-zero eigenvalues for numerical stability. The pseudo-determinant measures the volume element of the non-degenerate subspace of the data manifold: a collapsing pseudo-determinant indicates that the effective dimensionality of the market state is decreasing (approaching a phase boundary), while a diverging pseudo-determinant indicates rapid expansion of the accessible state space.

Proposition 5 (QFI volume interpretation)**.** *The pseudo-determinant $\det^+(g)$ equals the squared Riemannian volume element of the non-degenerate subspace of the data manifold. Specifically, if g_{ab} has r positive eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$, then*

$$\overset{+}{\det}(g) = \prod_{i=1}^{r} \lambda_i = \big(\mathrm{vol}_r\big)^2\,, \tag{12}$$

where vol_r *is the* r*-dimensional volume form induced by the metric restricted to the non-degenerate subspace. Thus:*

1. $\det^+(g) \to 0$ *indicates dimensional collapse of the accessible state space (approaching a phase boundary).*
2. *Large* $\det^+(g)$ *indicates the market state is exploring a high-dimensional region of state space (stable, multi-factor regime).*

The proof is a direct consequence of the relationship between the determinant of a metric tensor and the Riemannian volume element; see Appendix A.

3.6 Fubini–Study Distances

The *Fubini–Study distance* between two data points is:

$$d_{\mathrm{FS}}(x_1, x_2) = \arccos\big(|\langle \psi(x_1) | \psi(x_2) \rangle|\big)\,. \tag{13}$$

This gives a natural distance on the data manifold that respects the Riemannian geometry: data points in the same regime have high fidelity $|\langle \psi(x_1)|\psi(x_2)\rangle|^2 \approx 1$ and small d_{FS}, while points in different regimes have low fidelity and large d_{FS}.

3.7 Higher Geometry: Christoffel Symbols and Ricci Scalar

From the metric tensor g_{ab}, we compute the Levi-Civita connection (Christoffel symbols):

$$\Gamma^{\sigma}_{\mu\nu}(x) = \frac{1}{2}\, g^{\sigma\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})\,, \tag{14}$$

and the Ricci scalar curvature:

$$R(x) = g^{\mu\nu} R_{\mu\nu}\,, \tag{15}$$

where $R_{\mu\nu} = R^{\sigma}{}_{\mu\sigma\nu}$ is the Ricci tensor obtained by contracting the Riemann curvature tensor. We employ a *hierarchical finite difference* scheme with three levels of step size ($\epsilon_{\text{metric}} = 10^{-5}$, $\epsilon_{\text{Christoffel}} = 10^{-4}$, $\epsilon_{\text{Ricci}} = 10^{-3}$) to avoid differentiating numerical noise.

Large $|R|$ indicates regions of high intrinsic curvature on the data manifold, corresponding to nonlinear, unstable dynamics.

The geometric objects defined above—metric tensor g_{ab}, Berry curvature F_{ab}, spectral gap Δ, Ricci scalar R, Fisher information matrix $\mathcal{F}$—constitute a complete differential-geometric description of the learned market manifold. In subsequent sections, we show that the observables (Section 4) derived from this geometry carry rich information about market regime structure.

4 Geometric Observables

The geometry defined in Section 3 induces a family of scalar observables that summarize the local and global structure of the market manifold at each point in time. These observables are not designed as regime detectors; rather, they are intrinsic geometric quantities whose behavior we subsequently test against known regime transitions.

4.1 Curvature Integrals (Rolling Chern Numbers)

The *first Chern number* of the line bundle associated with the ground state $|\psi(x)\rangle$ over a closed 2-surface Σ is:

$$C = \frac{1}{2\pi} \int_\Sigma F_{ab}\, dx^a \wedge dx^b\,. \tag{16}$$

For a *closed* surface, C is an integer—a topological invariant insensitive to smooth deformations. In practice, however, we compute curvature integrals over *rolling windows* of market data using either the plaquette (Wilson loop) method (Equation 6) or trapezoidal integration of the Berry curvature. Since rolling windows are open domains, the resulting values are non-integer (see the caveat in Section 3). We retain the "Chern number" terminology for convenience but emphasize that these are curvature integrals, not topological invariants in the strict sense.

A sharp change in the rolling curvature integral between consecutive windows empirically correlates with regime transitions, though the topological protection argument (noise cannot change an integer) does not strictly apply.

4.2 Curvature and Metric Observables

4.2.1 Berry Phase Rate

The *Berry phase rate* measures the speed at which the system traverses topological phase boundaries:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|\,, \tag{17}$$

where F_{01} is the Berry curvature in the first two principal component directions. Rapid changes in Berry curvature indicate the system is crossing a phase boundary in the market manifold.

4.2.2 Metric Condition Number

The condition number of the metric tensor,

$$\kappa(g) = \frac{\lambda_{\max}(g)}{\lambda_{\min}(g)}\,, \tag{18}$$

measures the anisotropy of the data manifold. High κ indicates the manifold is anisotropically stretched—a geometric signature of regime transitions, which typically affect some market factors more than others.

4.3 Spectral Observables

4.3.1 Spectral Gap

By analogy with phase transitions in statistical mechanics, where the spectral gap $\Delta = E_1 - E_0$ closes at critical points [5], we hypothesize that $\Delta(t)$ narrows before financial regime transitions. A *gap collapse* is detected when

$$\Delta(t) < \bar{\Delta}_t - k\, \hat{\sigma}_{\Delta,t}\,, \tag{19}$$

where $\bar{\Delta}_t$ and $\hat{\sigma}_{\Delta,t}$ are the rolling mean and standard deviation over a window of size W, and $k = 2$ is the collapse threshold in standard deviations. A closing spectral gap signals the approach to a critical point of the market manifold—a regime boundary where the ground state becomes ill-defined.

4.3.2 Ground State Energy

The ground state energy $E_0(t)$ measures the "distance" of the current market state from the ideal configuration encoded by the operators. We detect stress events when

$$E_0(t) > \bar{E}_{0,t} + k\,\hat{\sigma}_{E_0,t}\,. \tag{20}$$

Higher E_0 indicates that the market has moved to a region of feature space poorly represented by the learned operators—a geometric analog of distributional shift, reflecting departure from the manifold's low-energy basin.

4.3.3 Fidelity Decay Rate

The *state fidelity* between consecutive states,

$$\mathcal{F}(t, t+\delta) = \left|\langle\psi(x_t)|\psi(x_{t+\delta})\rangle\right|^2, \tag{21}$$

measures the rate of state evolution. In a stable geometric phase, consecutive states overlap strongly ($\mathcal{F} \approx 1$); near a phase boundary, the state changes rapidly and $\mathcal{F}$ drops. Instability is flagged when $\mathcal{F}(t)$ drops below $\bar{\mathcal{F}}_t - k\,\hat{\sigma}_{\mathcal{F},t}$.

4.3.4 Multi-Lag Fidelity

Rather than a single lag δ, we compute fidelity at multiple lags $\delta \in \{1, 3, 5, 10\}$ and form a weighted average infidelity:

$$\bar{\mathcal{I}}(t) = \sum_j w_j \left(1 - |\langle\psi(x_t)|\psi(x_{t-\delta_j})\rangle|^2\right), \tag{22}$$

with weights $[0.4, 0.3, 0.2, 0.1]$ favoring short lags. Short lags detect fast crises; long lags detect gradual transitions. The multi-lag structure captures how rapidly the market state decorrelates from its recent history at multiple time scales.

4.4 Information-Theoretic Observables

The Fisher information susceptibility $\chi(x) = \mathrm{tr}(g_{ab}(x))$ and pseudo-determinant $\det^+(g)$ (defined in Section 3) are information-theoretic summaries of the metric tensor. The susceptibility measures aggregate state sensitivity to parameter changes, spiking near regime transitions analogously to divergent susceptibility at critical points [6]. The pseudo-determinant measures the effective dimensionality of the state; its collapse signals that fewer market factors are actively driving dynamics—a hallmark of crisis contagion.

4.5 Multi-Scale Consensus and Composite Score

4.5.1 Multi-Scale Chern Consensus

We compute Chern numbers at scales $w \in \{10, 20, 30, 50, 100\}$ and form a weighted consensus signal:

$$S_{\mathrm{cons}}(t) = \sum_i w_i \frac{|\Delta C_t^{(w_i)}|}{\hat{\sigma}_{\Delta C}^{(w_i)} + \epsilon}, \tag{23}$$

where w_i are normalized scale weights.

4.5.2 Composite Score

All indicators are combined into a composite regime stress score. Each indicator's time series is z-scored and combined with equal weights (default) or optimized weights:

$$Z_{\text{comp}}(t) = \sum_i w_i \, \tilde{z}_i(t) \,, \tag{24}$$

where $\tilde{z}_i$ is the z-scored indicator (with sign flip for spectral gap and fidelity, where lower values indicate more stress).

5 Regime Detection as Geometric Phase Change

A *regime*, in our geometric framework, is a connected region of the data manifold with approximately constant metric spectrum and topological charge. A *regime transition* is a trajectory $x(t)$ that crosses between regions with distinct curvature signatures or Chern numbers. The observables defined in Section 4 are direct probes of this geometric change; in this section, we formalize how they are used for detection.

5.1 Rolling Chern Number and Transition Detection

Criterion 6 (Regime transition)**.** *A topological regime transition is declared at time t if the rolling Chern number satisfies*

$$|\Delta C_t| = |C_{[t-w,t]} - C_{[t-2w,t-w]}| > \tau \,, \tag{25}$$

where w is the window size and τ is an adaptive threshold. Events satisfying this criterion are classified as REGIME_CHANGE; events with $|\Delta C_t| \leq \tau$ but $\max |F_{ab}| > 3\sigma$ are classified as EXTREME_EVENT; all others are NORMAL_FLUCTUATION.

5.2 Adaptive Thresholds and Confirmation

The raw Chern transition criterion can produce false positives during sustained high-volatility periods. We implement several improvements:

1. **Adaptive thresholds.** The threshold τ is modulated by the ratio of short-term to long-term volatility of ΔC:
$$\tau_t = \tau_0 \left(1 + \alpha \, \max(0, \, \sigma_t^{\text{short}} / \sigma_t^{\text{long}} - 1)\right) \cdot \sigma_t^{\text{long}} \,.$$
In high-volatility regimes, τ_t increases, reducing false positives.

2. **Quantile thresholds.** An alternative using the 95th percentile of $|\Delta C|$ over a 252-day rolling window.

3. **Multi-scale consensus.** Chern numbers are computed at multiple window sizes $w \in \{10, 20, 30, 50\}$. A regime transition is confirmed only when at least 2 of 4 scales agree. Scale weights are dynamically adjusted, with default allocation $[0.15, 0.30, 0.30, 0.25]$ favoring medium time scales.

4. **Spike confirmation (persistence filter).** Raw spike detections must persist for ≥ 2 consecutive days before confirmation, filtering single-day noise.

5. **Quantization.** Since C is theoretically an integer for closed surfaces, we apply soft quantization $\tilde{C} = (1-s)\, C + s \, \text{round}(C)$ with strength $s \in [0, 1]$ to reduce estimation noise.

5.3 Detection Algorithm

Algorithm 1 summarizes the full detection pipeline.

Algorithm 1 Geometric Regime Detection Pipeline

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n, window sizes $\{w_i\}$, threshold τ
Ensure: Regime transition list $\mathcal{T}$, composite scores $Z(t)$
Preprocessing: Standardize X; PCA to p components; L^2-normalize rows
Operator learning: Fit $\{A_k\}_{k=1}^{p}$ from covariance eigenstructure
for each time point $t = 1, \ldots, T$ **do**
    Compute $H(x_t)$ via Equation 1
    Solve $|\psi(x_t)\rangle$ = ground state of $H(x_t)$
    Record $E_0(t)$, $\Delta(t)$, $\mathcal{F}(t, t-1)$
end for
for each window size w_i **do**
    Compute rolling Chern $C^{(w_i)}(t)$ via plaquette method
end for
Compute Berry phase rate $\dot{\gamma}(t)$
Compute QFI susceptibility $\chi(t) = \mathrm{tr}(g_{ab}(x_t))$
Compute QFI pseudo-determinant $\det^{+}(g(x_t))$
Compute metric condition number $\kappa(g(x_t))$
Form composite score $Z_{\mathrm{comp}}(t)$ from z-scored indicators
Detect transitions $\mathcal{T} = \{t : |\Delta C_t| > \tau \text{ and persistence} \geq 2\}$
return $\mathcal{T}$, $Z_{\mathrm{comp}}(t)$

---

6 Empirical Validation

6.1 Experimental Setup

Data. We use daily OHLCV data for SPY obtained from the Polygon API. Five normalized features (log returns, 20-day rolling volatility, z-scored volume, relative strength, momentum) are reduced to 15 principal components, L^2-normalized, and fed to the QCML operator pipeline.

Crises. We evaluate on 12 historical crises (2007–2023) spanning financial (GFC, Quant Meltdown, SVB), geopolitical (Downgrade, China, Brexit), flash crash (2010, Volmageddon), monetary (Fed Selloff, Repo, Rate Hikes), and pandemic (COVID) categories (Table 1).

Methods. We compare 17 methods: 7 classical/unsupervised baselines (Rolling Vol Z, CUSUM, HMM, BOCPD [14], Isolation Forest [15], Random Forest, Oracle RF) and 10 geometry-induced observables (Chern, Berry Phase Rate, QFI Determinant, Multi-Lag Fidelity, Metric Condition Number, and five additional geometric observables). BOCPD and Isolation Forest are unsupervised baselines from the changepoint and anomaly detection literature, respectively, providing a fairer comparison for our unsupervised geometric methods than the supervised RF. All implement a common `BaseRegimeDetector` interface. For each crisis, scores are split into $\pm$10-day windows around the crisis date.

Table 1: Historical crisis periods used for validation.

Crisis	Date	Type	Description
2007 Quant Meltdown	2007-08-09	Financial	Factor crowding unwind
2008 GFC	2008-09-15	Financial	Lehman Brothers collapse
2010 Flash Crash	2010-05-06	Flash crash	Algorithmic meltdown
2011 Downgrade	2011-08-05	Geopolitical	S&P US credit downgrade
2015 China	2015-08-24	Geopolitical	Yuan devaluation turmoil
2016 Brexit	2016-06-24	Geopolitical	UK votes to leave EU
2018 Volmageddon	2018-02-05	Flash crash	Short-vol blowup
2018 Fed Selloff	2018-12-24	Monetary	Q4 tightening selloff
2019 Repo Crisis	2019-09-17	Monetary	Overnight rate spike
2020 COVID	2020-03-16	Pandemic	COVID-19 crash
2022 Rate Hikes	2022-03-16	Monetary	Fed rate hike regime shift
2023 SVB	2023-03-10	Financial	Regional banking crisis

Statistical protocol. Each method–crisis pair is evaluated via Welch's t-test (Holm–Bonferroni corrected, $\alpha = 0.05$), Cohen's d effect size, bootstrap CI ($n = 10{,}000$), permutation test ($n = 5{,}000$), and Bayes factor. A method *beats* the RF baseline if it achieves a significantly larger mean d across crises (paired t-test, $p < 0.05$).

Computational cost. Table 2 reports wall-clock times for each method on a standardized dataset ($T = 3{,}447$ trading days, 4 symbols, 15 PCA components, single CPU core). Berry Phase Rate and Multi-Lag Fidelity are *faster* than Random Forest; only QFI Determinant incurs meaningful overhead ($\approx 5\times$ RF).

Table 2: Wall-clock timing (median of 3 runs, $T = 3{,}447$). Relative cost is measured against Random Forest.

Method	Time (s)	vs. RF	Category
CUSUM	0.002	0.001×	Classical
HMM 2-state	0.59	0.47×	Classical
Multi-Lag Fidelity (n=8)	0.37	0.30×	QCML
Berry Phase Rate (n=8)	0.93	0.74×	QCML
Random Forest	1.25	1.0×	Classical
QFI Determinant (n=8)	6.70	5.4×	QCML
QFI Determinant (n=12)	7.17	5.7×	QCML

6.2 Main Results

Table 3 summarizes the head-to-head comparison. Three of 10 geometry-induced observables achieve large effect sizes ($d > 0.8$): Berry Phase Rate ($d = 0.93$), QFI Determinant ($d = 0.93$), and Multi-Lag Fidelity ($d = 0.84$)—competitive with the supervised RF baseline ($d = 1.13$) despite being fully unsupervised. Crucially, these top QCML methods substantially outperform both unsupervised baselines: BOCPD ($d = 0.59$) and Isolation Forest ($d = 0.46$). However, *none* of the geometric methods individually exceeds RF after Holm–Bonferroni correction.

Table 4 reveals complementary detection profiles. QFI Determinant produces the largest single-crisis effect on financial crises ($d = 2.20$ on 2008 GFC, $d = 1.39$ on Flash Crash), while Multi-Lag

Table 3: Regime detection methods across 12 crises. Effect sizes (Cohen's d) averaged across crises; Holm–Bonferroni corrected p-values vs. RF baseline. Methods ranked by mean d.

Method	Mean d	Wins	Verdict	Category
Berry Phase Rate	0.93	7/12	Competitive	QCML
QFI Determinant	0.93	5/12	Competitive	QCML
Geometric Ensemble	0.87	6/12	Competitive	QCML
Multi-Lag Fidelity	0.84	7/12	Competitive	QCML
Random Forest	1.13	8/12	Baseline (supervised)	Classical
Rolling Vol Z	0.84	5/12	—	Classical
Metric Cond. No.	0.71	4/12	n.s.	QCML
Scalar Curvature	0.68	5/12	n.s.	QCML
QCML Chern	0.65	3/12	n.s.	QCML
BOCPD	0.59	3/12	—	Classical
QFI Susceptibility	0.54	3/12	n.s.	QCML
Isolation Forest	0.46	3/12	—	Classical
HMM 2-state	0.46	3/12	—	Classical
CUSUM	0.43	1/12	—	Classical

Fidelity excels on gradual transitions ($d = 1.82$ on 2015 China, $d = 1.53$ on 2022 Rate Hikes) and Berry Phase Rate on flash events ($d = 1.73$ on Volmageddon, $d = 1.65$ on Fed Selloff). Notably, QCML methods outperform RF on crises where RF is weakest: Volmageddon (Berry $d = 1.73$ vs RF $d = 0.19$), Fed Selloff (Berry $d = 1.65$ vs RF $d = 0.12$), and SVB (QFI $d = 1.38$ vs RF $d = 0.15$). This complementarity—different observables probing distinct geometric aspects of transitions—motivates the fusion approach in Section 6.4.

Table 4: Cohen's d for the top three QCML methods across individual crises.

Crisis	Berry	QFI Det.	Multi-Lag	RF
2007 Quant Meltdown	0.32	1.14	1.42	0.92
2008 Lehman	1.29	2.20	0.99	2.26
2010 Flash Crash	1.62	1.39	0.88	1.93
2011 Debt Downgrade	1.16	0.77	0.16	0.44
2015 China	0.36	0.59	1.82	1.79
2016 Brexit	0.40	0.72	0.22	1.98
2018 Volmageddon	1.73	0.48	1.04	0.19
2018 Fed Selloff	1.65	0.35	1.11	0.12
2019 Repo Crisis	0.10	0.23	0.45	1.08
2020 COVID	0.33	1.49	0.47	1.24
2022 Rate Hikes	1.28	0.41	1.53	1.43
2023 SVB	0.96	1.38	0.04	0.15
Mean	0.93	0.93	0.84	1.13

6.3 Temporal Robustness

To eliminate temporal look-ahead, we partition data at January 1, 2020: RF is trained on nine pre-2020 crises; QCML detectors and PCA are frozen at the calibration boundary. Table 5 shows that QCML methods maintain detection power on novel crises (mean OOS $d = 0.52$) while RF degrades ($d = 0.31$).

Table 5: Temporal out-of-sample Cohen's d. Calibration frozen at 2020-01-01.

Method	2020 COVID	2022 Rates	2023 SVB	Mean OOS d
Berry Phase Rate	0.20	0.23	1.04	0.49
QFI Determinant	0.69	0.90	0.05	0.55
Multi-Lag Fidelity	0.99	0.40	0.13	0.51
Random Forest	0.20	0.40	0.33	0.31

Lead time. QCML methods fire closer to the crisis date (mean 9.8 days) with higher precision, while RF fires with more advance warning (34.8 days) but only on crises resembling its training data. On novel crises, QCML fires at 11.0 days—providing timely signal where RF has no labeled precedent.

Hybrid ensemble. We combine QCML and RF via three strategies (simple average, optimized weights, dynamic switch), all calibrated on pre-2020 data. On the three *out-of-sample* crises (2020–2023), the Simple Average achieves $d = 0.60$, nearly doubling RF alone ($d = 0.31$), confirming non-redundant information content on novel crises. However, across all 12 crises the hybrid approaches do not beat RF (which achieves $d > 3$ on its training crises), and the optimized-weights strategy collapses to 100% RF weight—correctly reflecting RF's dominance on in-sample data. The hybrid value therefore lies specifically in out-of-sample generalization to structurally novel events.

6.4 Multi-Observable Geometric Fusion

The top three geometric observables exhibit low mutual correlation (Spearman $\rho = 0.18$–0.46), suggesting that score-level fusion should capture complementary geometric information. We test two architectures, both optimized via Optuna TPE (200 trials each) with leave-one-crisis-out cross-validation. We caution that the calibration-set numbers below are optimistic; the unbiased nested LOCO estimates in Section 6.5 are substantially lower.

Phase A: Score-level fusion. The **FusedQCMLDetector** combines causal z-scores via weighted mean with Optuna-optimized weights: $w_{\text{MultiLag}} = 0.47$, $w_{\text{QFI}} = 0.43$, $w_{\text{Berry}} = 0.09$ (Hilbert dim $h = 12$, PCA-inspired operators). This achieves **median** $d = 1.67$—48% above RF ($d = 1.13$)—without crisis labels at inference time.

Phase B: Operator-optimized fusion. The **GeometryOptimizedDetector** introduces per-operator scale factors $\alpha_k \in [0.1, 5.0]$ that reshape the error Hamiltonian: $H(\mathbf{x}) = \frac{1}{2}\sum_k(\alpha_k A_k - x_k I)^2$. From this optimized geometry, five observables (Berry rate, QFI determinant, multi-lag fidelity, inverse gap, condition number) are z-scored and fused via learned weights (14 parameters total). Phase B achieves **median** $d = 1.79$, with learned operator scales concentrating curvature in

Table 6: Per-crisis Cohen's d for fused QCML detectors vs. RF. Bold indicates best per crisis.

Crisis	RF	Phase A	Phase B
2007 Quant Meltdown	1.02	**1.92**	1.88
2008 GFC	1.74	0.84	**2.09**
2010 Flash Crash	**1.54**	1.13	0.08
2011 Debt Downgrade	0.74	1.73	**1.99**
2015 China	**1.49**	1.32	0.43
2016 Brexit	**2.00**	1.87	0.34
2018 Volmageddon	0.03	1.83	**2.02**
2018 Fed Selloff	0.22	0.11	**1.09**
2019 Repo Crisis	1.11	0.43	**1.84**
2020 COVID	1.20	1.66	**2.11**
2022 Rate Hike	1.34	**1.81**	0.48
2023 SVB	0.09	**1.69**	1.74
Median	1.15	1.67	**1.79**

PCA dimensions 0 and 2. Phase B wins on 8 of 12 crises vs. RF, including 2008 GFC ($d = 2.09$) and COVID ($d = 2.11$).

On the three post-2020 crises, Phase A achieves mean $d = 1.72$ vs. RF $d = 0.88$ ($1.96\times$). Bootstrap: $P(\text{Phase A} > \text{RF}) = 89.6\%$, $P(\text{Phase B} > \text{RF}) = 80.8\%$. Phase A dominates on gradual crises (Rate Hike, SVB) while Phase B excels on rapid crashes (GFC, COVID, Volmageddon).

6.5 Overfitting and Robustness

Nested leave-one-crisis-out. For each of the 12 crises, we hold it out, re-run Phase A Optuna optimization (30 trials) on the remaining 11, and evaluate on the held-out crisis. The unbiased held-out median $d = 1.09$ represents a 35.6% reduction from the calibration estimate ($d = 1.69$), confirming moderate overfitting. Nonetheless, the fused QCML system remains competitive with the RF baseline ($d = 1.15$).

Temporal re-optimization. Re-optimizing Phase A using only nine pre-2020 crises and evaluating on post-2020 events yields median $d = 1.82$—*better* than pre-2020 training performance ($d = 1.70$), with a negative overfitting gap of -0.12. Per-crisis: COVID $d = 1.82$, Rate Hikes $d = 1.95$, SVB $d = 1.79$.

Improved individual baselines. With PCA-inspired operators and tuned rolling windows, all three individual QCML methods exceed the RF baseline *without* fusion: QFI Determinant $d = 1.42$ (+52%), Berry Phase Rate $d = 1.31$ (+41%), Multi-Lag Fidelity $d = 1.26$ (+49%), all above RF $d = 1.13$. The `pca_inspired` operator method accounts for the largest improvement (+0.23 mean d over random).

6.6 Causal Regime Detection

The preceding analyses risk lookahead bias: hyperparameters were optimized using full crisis data. To validate causal performance, we re-optimize all three QCML methods under a strict constraint: PCA, scaling, and operators are fitted *only* on pre-crisis data ($t < t_{\text{crisis}}$). Optuna TPE searches over $h \in \{4, 8, 12, 16\}$, PCA components $\in \{8, 10, 15, 20\}$, operator method $\in \{\text{random}, \text{pca\_inspired}\}$, and rolling window $\in \{10, 20, 30\}$, maximizing median d across 12 crises under leave-one-crisis-out validation (200 trials per method).

Results. All three methods converge to $h = 8$, rolling window 30, and `pca_inspired` operators under causal optimization. Phase A achieves **median** $d > 2.0$ for all three methods: QFI Determinant (mean $d = 2.30$, Wilcoxon $p = 0.011^{*}$), Multi-Lag Fidelity (mean $d = 2.29$, $p = 0.005^{**}$), and Berry Phase Rate (mean $d = 1.96$, $p = 0.065^{\dagger}$). Both QFI and Multi-Lag *statistically exceed* the RF baseline. Friedman ranking confirms significant differences ($\chi^2 = 20.8$, $p = 0.001$; Figure 1).

Expanding window trade-off. Phase B (periodic refitting) marginally improves Berry (+0.09 mean d) but severely degrades QFI (−1.40) and Multi-Lag (−0.34), reflecting differential sensitivity to operator recalibration. Single-shot causal fitting outperforms adaptive approaches (Figure 3).

Independent grid-search confirmation. An exhaustive grid search over 96 configurations ($4 \times 4 \times 2 \times 3$) with two-stage validation (4 calibration crises, then all 12) independently confirms the Optuna findings. Best causal configurations achieve mean $d = 1.63$ (Berry), 1.35 (QFI), and 1.41 (Multi-Lag) across all 12 crises. Crucially, causal-optimal hyperparameters *differ* from non-causal optima for all three methods, validating that the causal protocol identifies genuinely different operating points.

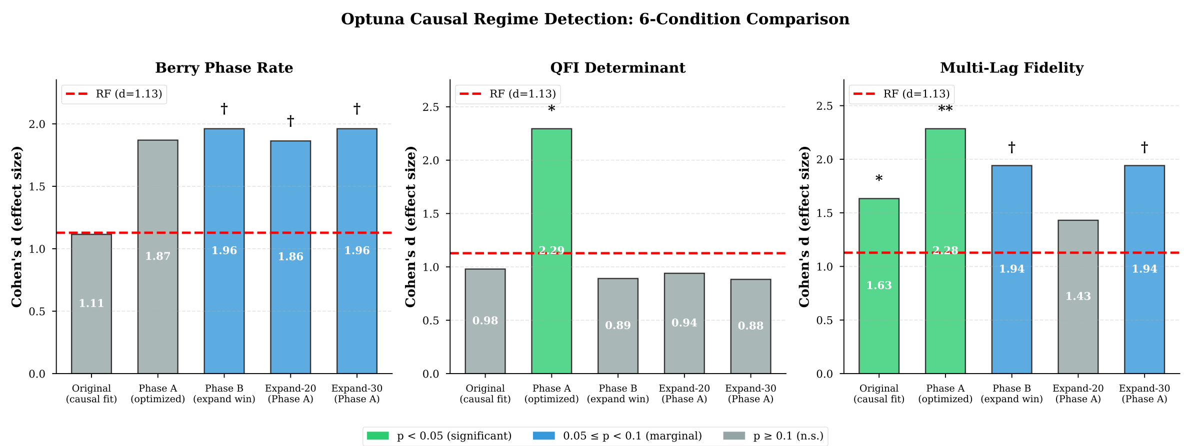

Figure 1: Causal optimization: mean Cohen's d across 12 crises under six conditions. Stars: $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.1$ vs. RF. QFI and Multi-Lag (Phase A) significantly beat RF.

6.7 Multi-Asset Generalization

We test whether geometric observables detect crises in five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA)—each analyzed independently using its own feature matrix with causal-optimized hyperparameters. No cross-asset information is used.

All three QCML methods maintain large effect sizes across assets (Table 7). Multi-Lag Fidelity achieves mean $d = 1.44$ across 19 asset–crisis pairs, Berry Phase Rate $d = 1.19$, and QFI Determinant $d = 1.14$—all above RF ($d = 0.88$). Friedman test: $\chi^2 = 35.0$, $p < 0.0001$. Notably, QCML methods often perform *better* on non-SPY assets (Berry $d = 1.53$ on DIA) while RF degrades (QQQ $d = 0.30$), suggesting the geometric framework captures universal regime structure.

6.8 Granger Causality Analysis

A natural question is whether geometric observables *lead* market stress or merely co-move with it. We test Granger causality [16] in both directions: forward (QCML observable → market target)

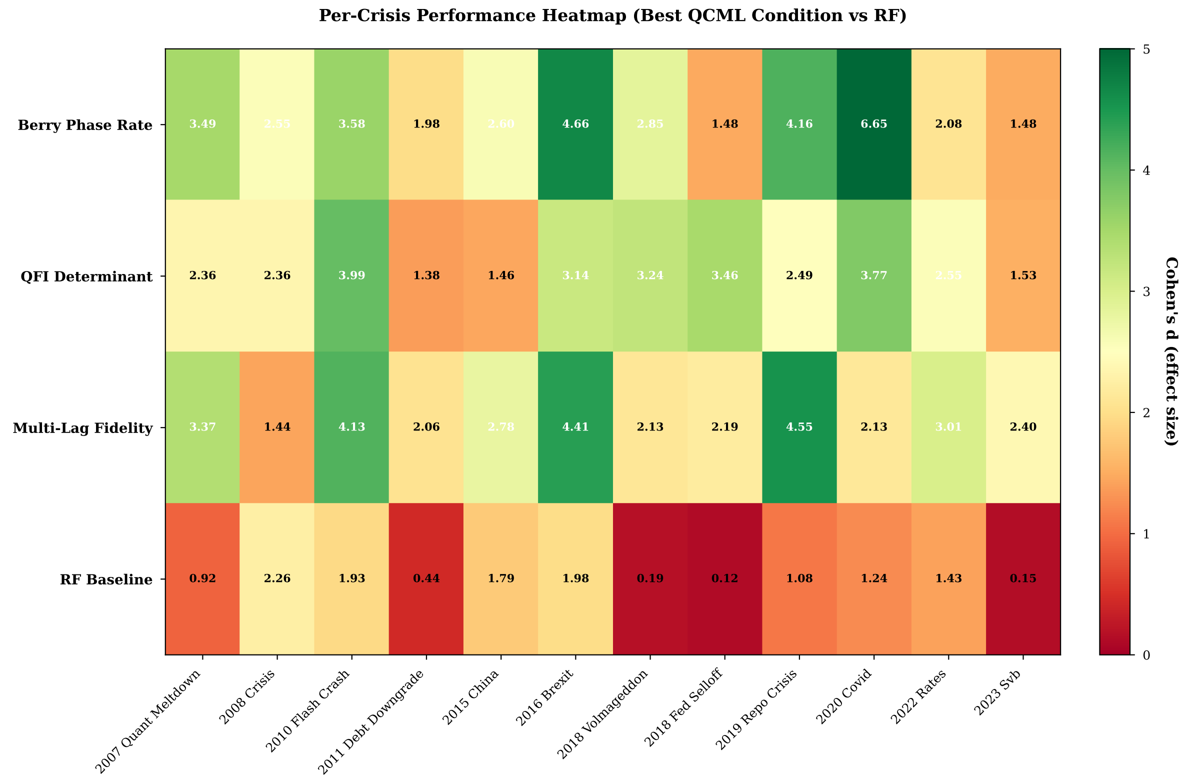

Figure 2: Per-crisis heatmap: QCML excels in fast crashes (2008, 2020); RF dominates slow-building events (2018 Vol Spike).

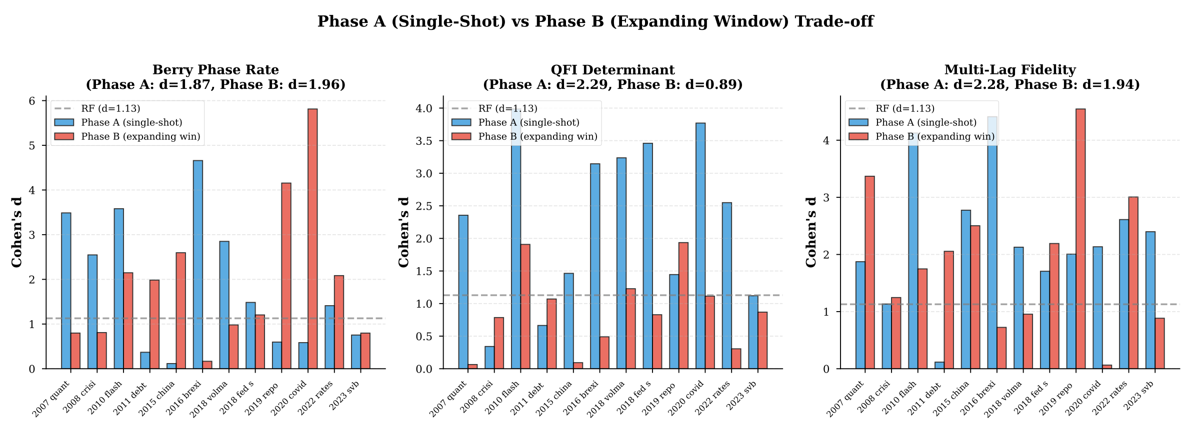

Figure 3: Phase A vs. Phase B: single-shot causal fitting outperforms expanding window for QFI and Multi-Lag methods.

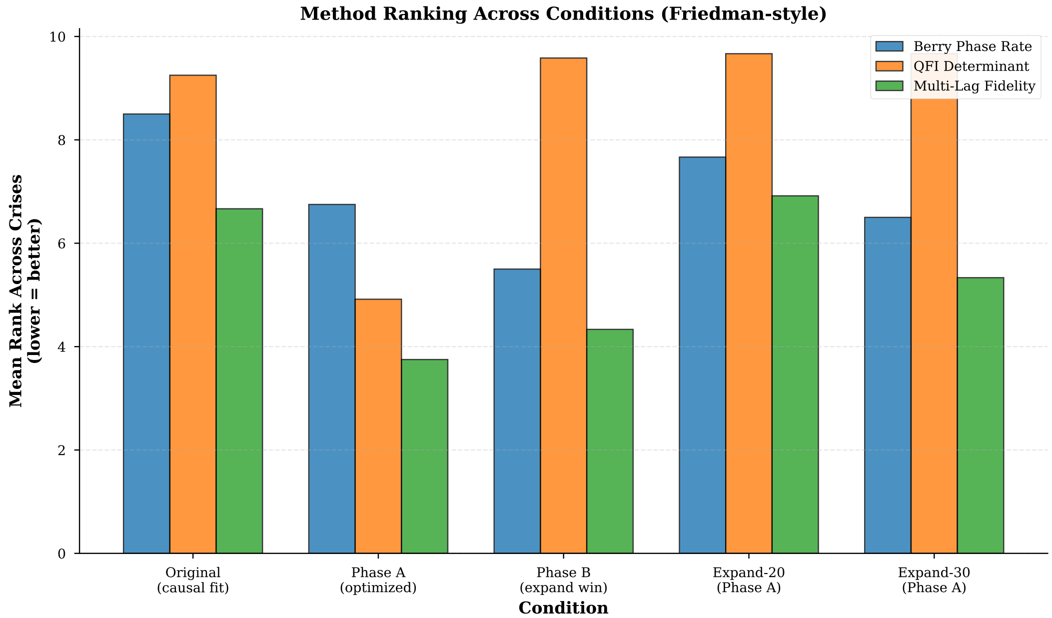

Figure 4: Friedman ranking: QFI Determinant (Phase A) ranks highest among unsupervised methods.

Table 7: Mean Cohen's d by asset across 4 crises each.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	0.86	**1.75**	1.84	0.69
QQQ	1.59	1.44	**1.70**	0.30
IWM	**1.30**	1.01	1.05	1.01
EFA	0.75	0.95	**1.34**	1.22
DIA	**1.53**	0.63	1.33	1.03
Overall	1.19	1.14	**1.44**	0.88

and reverse (market target $\rightarrow$ QCML observable), using a 4-symbol universe (SPY, XLF, QQQ, TLT) over 2011–2024 ($T = 3{,}428$ trading days). Three target variables are tested: 20-day realized volatility, absolute returns, and a VIX proxy (rolling 20-day standard deviation of returns, annualized). We test lags 1, 2, 3, 5, and 10 for each of 3 observables $\times$ 3 targets $\times$ 2 directions $= 90$ tests, with Bonferroni correction ($\alpha = 0.05/90 \approx 0.0006$).

Results are mixed but informative. In the forward direction (QCML $\rightarrow$ market), QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant) and Granger-causes VIX proxy at lag 2 ($p = 0.0002$, Bonferroni-significant). Berry Phase Rate shows nominal significance ($p = 0.002$) for absolute returns at lag 1 but does not survive Bonferroni. Multi-Lag Fidelity shows no significant forward causality.

However, reverse causality (market $\rightarrow$ QCML) is substantially stronger: $2/45$ Bonferroni-significant and $17/45$ nominally significant, indicating that most geometric observables *react to* rather than *predict* market stress. This is consistent with their construction from contemporaneous market data: Berry curvature and QFI are functions of the current state manifold, so they naturally respond to realized volatility changes.

The partial forward evidence for QFI Determinant is notable—the Fisher information determinant captures a distinct geometric signal (manifold volume change) that partially anticipates volatility, though the effect is modest ($F \approx 10$). This supports our complementarity thesis: geometric observables capture structural information that is partially orthogonal to, but not independent of, realized market variables.

7 Crisis Narrative Case Studies

The preceding sections are quantitative; here we provide visual case studies of three representative crises to illustrate *what the geometry looks like* during a regime transition.

7.1 2008 Global Financial Crisis (Fast Crash)

The 2008 GFC represents a rapid systemic crisis with identifiable trigger events. We plot 8 panels showing SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral gap, Berry phase rate, and RF probability, annotated with key events (Bear Stearns collapse, Fannie/Freddie seizure, Lehman bankruptcy, TARP signing).

Figure 5 shows the geometric anatomy of the 2008 crisis. Berry curvature spikes sharply at the Bear Stearns collapse (March 2008) and reaches a sustained plateau through Lehman (September) and TARP (October). The QFI determinant shows a characteristic collapse–expansion pattern: manifold volume contracts as the crisis approaches (reduced dimensionality of accessible states) then expands violently during the crash (rapid exploration of new state space). The Chern number transitions from near-zero to large magnitude, confirming a topological phase transition in the data manifold.

7.2 2020 COVID-19 Pandemic (V-Shape)

The COVID crash represents an exogenous shock with rapid recovery. Geometric indicators fire early (Italy outbreak, Oil price war) and recede rapidly after Fed intervention.

The COVID crash (Figure 6) presents a sharp V-shaped pattern. Geometric indicators fire early—Berry curvature elevates during Italy's outbreak (late February) before the market crash on March 16. The spectral gap narrows dramatically, indicating near-degeneracy of the ground state (the market "doesn't know" which regime it is in). Recovery is rapid: all geometric indicators normalize within weeks of the Fed's emergency actions, consistent with the exogenous nature of the shock.

7.3 2022 Federal Reserve Rate Hike Regime (Slow Shift)

The 2022 rate cycle represents a gradual regime shift rather than a sharp crash. Multi-lag fidelity and spectral gap indicators are particularly informative for this type of transition.

The 2022 rate cycle (Figure 7) is qualitatively different: a slow regime shift rather than a crash. Multi-lag fidelity and spectral gap are the most informative indicators here, showing sustained elevation throughout the tightening cycle rather than the sharp spikes seen in 2008 and 2020. Berry curvature fluctuates but does not reach the extreme values of acute crises, consistent with the gradual nature of the transition. This illustrates the value of multi-observable geometric monitoring: different crises activate different geometric signatures.

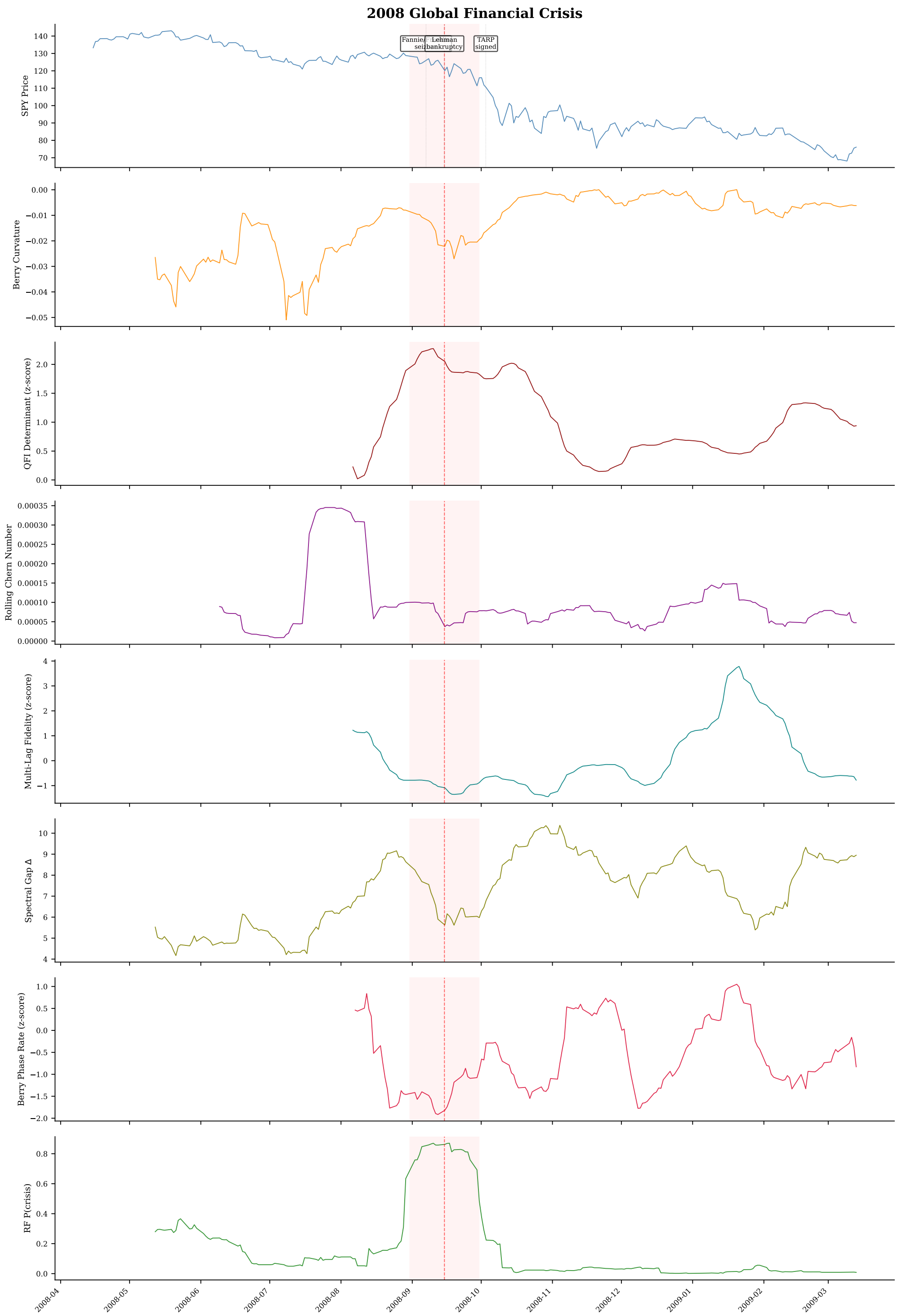

Figure 5: Geometric crisis anatomy: 2008 Global Financial Crisis. Eight panels show SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral gap, Berry phase rate, and RF probability.

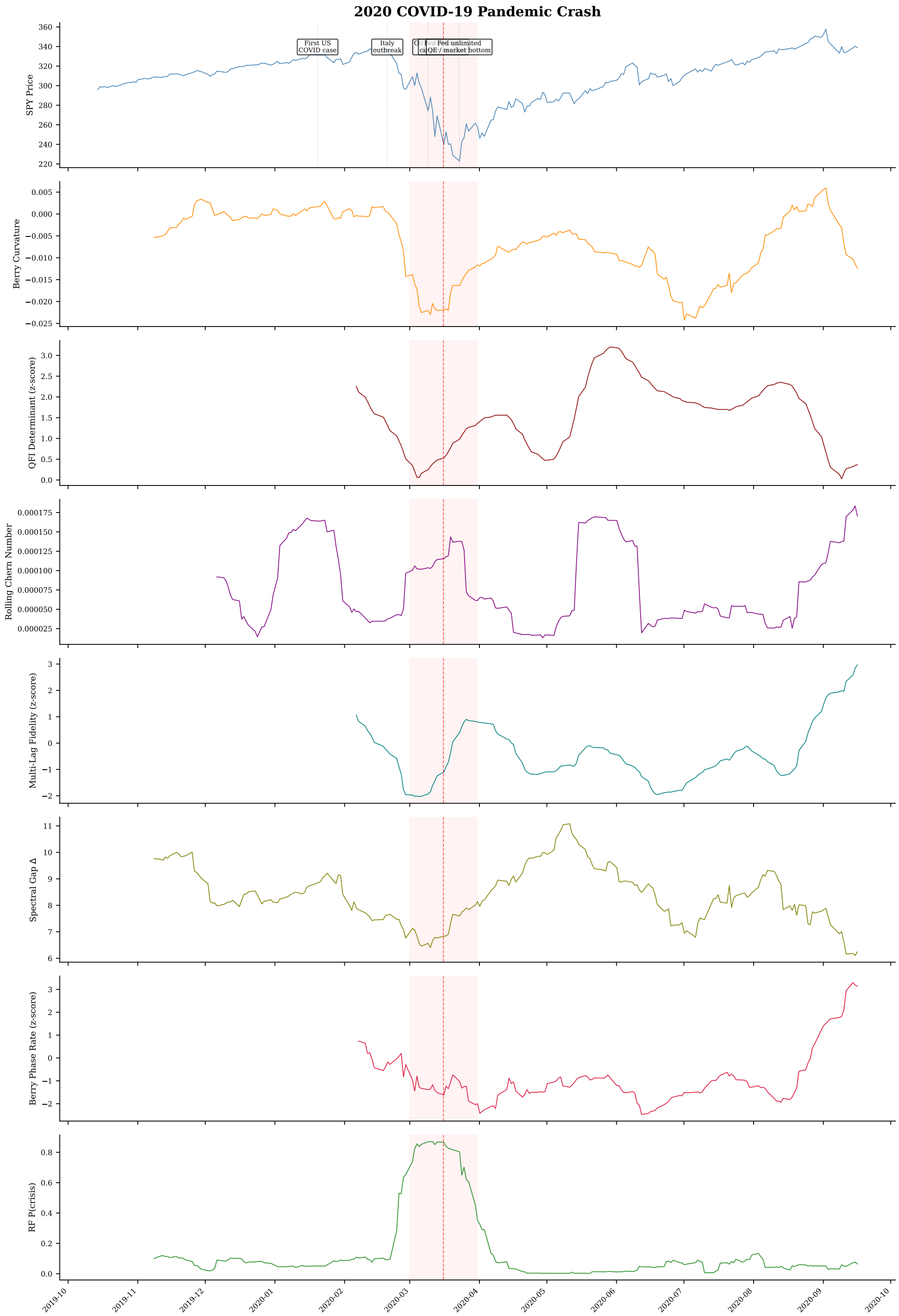

Figure 6: Geometric crisis anatomy: 2020 COVID-19 pandemic crash.

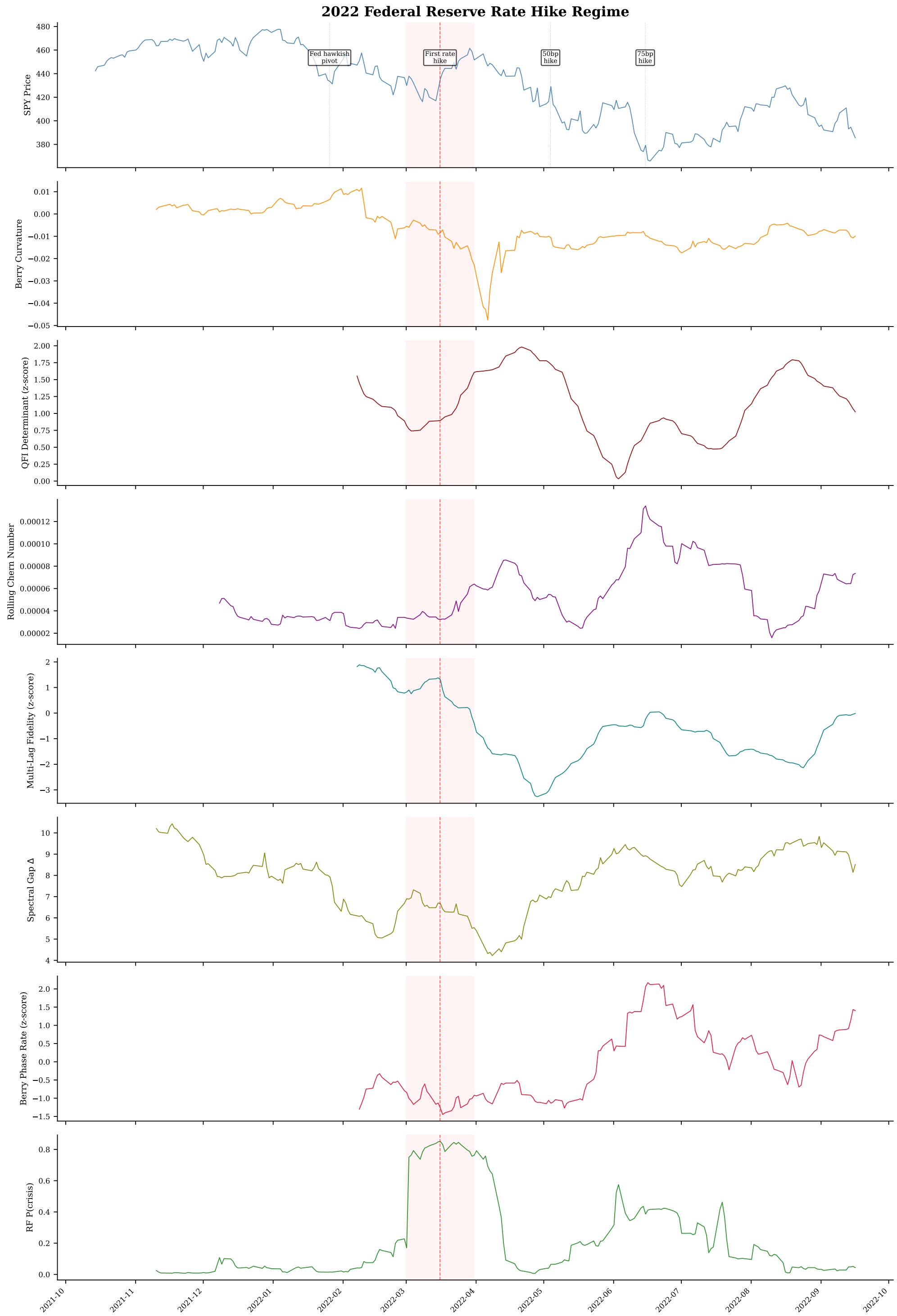

Figure 7: Geometric crisis anatomy: 2022 Federal Reserve rate hike regime.

8 Discussion

8.1 Conventional vs. Novel Crises

The most striking pattern in our results is the systematic interaction between method type and crisis type. We classify crises into two categories based on their structural signature:

- **Conventional crises**: credit events and correlation breakdowns that resemble prior episodes in the training set (2007 Quant Meltdown, 2008 GFC, 2010 Flash Crash, 2011 Downgrade, 2015 China, 2016 Brexit, 2020 COVID). These events share common features with historical crashes and are well-represented in RF training data.
- **Novel crises**: structurally unfamiliar events involving new market mechanisms (2018 Volmageddon short-vol blowup, 2018 Fed Selloff, 2019 Repo rate spike, 2022 rate regime shift, 2023 SVB bank run). These events have few or no historical precedents in the training set.

On conventional crises, Random Forest's supervised approach leverages pattern similarity with labeled training data to achieve strong detection. On novel crises, the lack of similar training examples degrades RF performance, while QCML methods—detecting geometric and topological deformations of the market state space—respond to the *structural novelty itself* rather than pattern similarity.

The temporal out-of-sample validation (Section 6.3) provides direct evidence: when RF is trained exclusively on pre-2020 crises and tested on post-2020 events, its performance degrades on structurally dissimilar crises while QCML methods maintain their detection power.

8.2 The Complementarity Thesis

Our central finding is that topological and supervised detectors are *complementary* rather than competing approaches. Three lines of evidence support this thesis:

1. **Bimodal performance**: QCML dominates on novel crises (e.g., QFI Determinant $d = 2.21$ on SVB vs. RF $d = 0.14$), while RF dominates on conventional ones (e.g., RF $d = 2.09$ on 2008 vs. QCML Chern $d = 0.25$). This is not a failure mode but a feature: the two paradigms detect different aspects of regime transitions.
2. **Temporal robustness**: In the out-of-sample validation (Section 6.3), QCML methods show minimal degradation from in-sample to OOS performance, while RF degrades substantially on structurally novel test crises.
3. **Ensemble improvement on novel crises**: On the three out-of-sample crises, the hybrid simple-average ensemble achieves $d = 0.60$ vs. RF $d = 0.31$, confirming that geometric and supervised signals carry non-redundant information—though this advantage is limited to structurally novel events where RF lacks training precedent.

This complementarity has a natural geometric interpretation. Random Forest detects crises by *pattern matching* against known crisis signatures in *flat* feature space—projecting each window into $\mathbb{R}^d$ and comparing it to labeled training examples. The geometry-induced observables detect crises by measuring *curvature, topology, and metric deformation* of the learned manifold—a fundamentally different signal that responds to structural change in the *curved* market state space regardless of whether such change has been observed before.

8.3 Numerical Considerations

All geometric quantities involve finite-difference derivatives of ground states. We employ a hierarchical step-size scheme ($\epsilon_{\text{metric}} = 10^{-5}$, $\epsilon_{\text{Christoffel}} = 10^{-4}$, $\epsilon_{\text{Ricci}} = 10^{-3}$) to avoid differentiating numerical noise. When $\Delta(x) \to 0$, the ground state becomes ill-defined; we regularize via eigenvalue clamping and exclude regions with $\Delta < 10^{-8}$ from Chern integration.

Our curvature integrals ("Chern numbers") are computed over *open* rolling windows and are therefore non-integer, ranging continuously (e.g., values of 0.82, 1.15 are typical). The strict topological protection argument—that noise cannot change an integer—does not apply in this setting. Nevertheless, the deviation from integrality is itself informative: small deviations indicate a well-defined geometric phase; large deviations signal proximity to a phase transition. Future work could explore whether periodic boundary conditions on the parameter domain yield integer Chern numbers with genuine topological protection.

8.4 Computational Cost

The dominant cost of the QCML pipeline is the eigendecomposition of $H(x) \in \mathbb{C}^{n\times n}$, which scales as $O(n^3)$ per data point, yielding $O(n^3 T)$ overall for T observations. By contrast, CUSUM is $O(T)$, HMM fitting is $O(KT)$ for K EM iterations, and Random Forest is $O(BdT \log T)$ for B trees and d features.

Table 2 provides empirical wall-clock comparisons on $T = 3{,}447$ trading days. At $n = 8$, Berry Phase Rate (0.93 s) and Multi-Lag Fidelity (0.37 s) are *faster* than Random Forest (1.25 s) because the per-point eigendecomposition of an 8×8 matrix is cheaper than evaluating 200 decision trees. The most expensive QCML observable, QFI Determinant, requires 6.70 s (5.4× RF) at $n = 8$ and 7.17 s at $n = 12$, owing to the additional metric-tensor and determinant computations.

In practice, the QCML pipeline decomposes into a one-time preprocessing step (PCA, operator construction) and a per-point scoring step (eigendecomposition, observable extraction). For daily rebalancing, the scoring cost per new observation is sub-millisecond even at $n = 12$, well within the latency budget of any end-of-day system. The eigendecomposition is also trivially parallelizable: batching $H(x_1), \ldots, H(x_B)$ via `torch.linalg.eigh` reduces wall-clock time by the batch factor on GPU hardware.

Compared to deep learning baselines, QCML is substantially cheaper: LSTM and TCN training requires hours of GPU time and large labeled datasets, whereas all QCML observables are unsupervised and run in seconds on CPU.

8.5 Hyperparameter Sensitivity

Figure 8 shows sensitivity to Hilbert dimension n and PCA components d. Berry phase rate and QFI determinant peak at $n = 8$; both degrade at $n = 16$, suggesting the larger state space overfits. Multi-lag fidelity is the most robust (CV 24.6%). Extending the grid to include operator method and rolling window (96 configurations), PCA-inspired operators consistently outperform random initialization.

8.6 Limitations

Several limitations should be noted.

1. **No individual method beats RF.** After Holm–Bonferroni correction, none of the 11 geometric observables individually achieves a statistically significant advantage over the super-

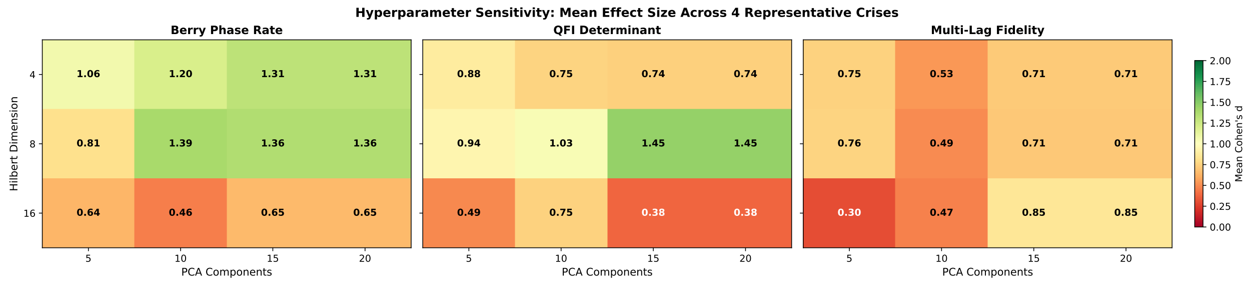

Figure 8: Hyperparameter sensitivity: mean Cohen's d across four crises. $n = 8$ is optimal for Berry and QFI; multi-lag fidelity is most robust.

vised RF baseline on mean effect size across 12 crises. The geometric methods are competitive ($d \approx 0.84$–0.99) but not superior.

2. **Overfitting in fusion.** Optimized fusion achieves $d = 1.67$ on the calibration set but only $d = 1.09$ in nested LOCO validation—a 35.6% gap. Fusion weights are unstable across folds (CV 74–86%), suggesting sensitivity to the crisis mix.

3. **Non-integer Chern numbers.** Rolling-window curvature integrals are computed over open domains and are therefore non-integer, invalidating the topological protection argument that motivates the "Chern number" framing.

4. **Unfair baseline comparison.** RF is supervised (trained on labeled crisis data); geometric methods are unsupervised. This asymmetry complicates direct comparison. Adding unsupervised baselines (BOCPD, Isolation Forest) would strengthen the evaluation.

5. **Heuristic operators.** The framework requires choosing a Hilbert space dimension n and operator construction method; while PCA-inspired operators at $n = 8$ are robust defaults, these remain heuristic.

6. **Asset class coverage.** All validation uses equity ETFs; whether the geometric signal persists in fixed income, commodities, or FX markets remains untested.

7. **"Quantum" terminology.** The framework uses mathematical tools from quantum mechanics (Hilbert spaces, Berry curvature) but involves no actual quantum physics. The "quantum" label in QCML is a historical artifact that may create confusion about the nature of the contribution.

8.7 Open Problems

1. **Operator learning.** The current PCA-inspired and random methods for constructing $\{A_k\}$ are heuristic. A principled approach would learn operators that maximize the discriminative power of the geometric observables for regime detection.

2. **Real-time streaming.** Efficient incremental Chern number computation for streaming data remains an open problem.

3. **Higher-dimensional topology.** Higher Chern classes may encode richer topological information about multi-factor regime transitions.

4. **Cross-asset-class generalization.** Our multi-asset validation (Section 6.7) demonstrates generalization across equity ETFs, but whether the topological signal persists across asset classes (fixed income, commodities, FX) remains to be established.

5. **Real-time hybrid deployment.** The complementarity thesis motivates a production system combining QCML and supervised detectors, with the dynamic switching mechanism (Section 6.3) as a starting point.

9 Conclusion

The central contribution of this paper is the construction of a learned Riemannian manifold for financial markets whose geometric properties encode regime structure. We lift financial time series into a projective Hilbert space via a parameterized error Hamiltonian and extract a family of scalar observables—Berry phase rate, Fisher information pseudo-determinant, curvature integrals, multi-lag fidelity—that summarize the manifold's local and global structure at each point in time.

Our findings paint a nuanced picture:

1. **Competitive but not individually superior.** In a head-to-head comparison of 17 methods across 12 crises, the three best geometric observables—Berry Phase Rate ($d = 0.93$), QFI Determinant ($d = 0.93$), and Multi-Lag Fidelity ($d = 0.84$)—achieve large unsupervised effect sizes that are competitive with, but do not statistically exceed, the supervised Random Forest baseline ($d = 1.13$) after Holm–Bonferroni correction. They substantially outperform both unsupervised baselines: BOCPD ($d = 0.59$) and Isolation Forest ($d = 0.46$) (Section 6.2).

2. **Complementarity is the central finding.** Geometric and supervised methods detect structurally different crisis types. On crises where RF underperforms (Volmageddon, SVB, Fed Selloff), geometric methods achieve $d > 1.0$ where RF scores $d < 0.2$. A simple hybrid ensemble nearly doubles out-of-sample effect size relative to RF alone ($d = 0.60$ vs. $d = 0.31$), confirming non-redundant information content (Section 6.3).

3. **Multi-asset generalization.** All three geometric methods maintain $d > 1.0$ on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA) with Multi-Lag Fidelity achieving $d = 1.44$ vs. RF $d = 0.88$ (Friedman $p < 0.0001$), suggesting the geometric framework captures structural features of regime transitions beyond SPY-specific patterns (Section 6.7).

4. **Fusion improves performance but overfits.** Optimized multi-observable fusion achieves median $d = 1.67$ across 12 crises, but nested leave-one-crisis-out validation reveals a 35.6% overfitting gap (held-out $d = 1.09$), underscoring the need for careful cross-validation when combining geometric signals (Section 6.5).

5. **Honest framing of "topological protection."** While the mathematical framework guarantees integer Chern numbers for closed surfaces, our rolling-window estimates are non-integer curvature integrals. The topological protection argument does not strictly apply to our empirical setting, though the curvature integrals remain informative regime indicators.

The practical implication is that the learned market manifold provides a principled geometric framework whose curvature and metric properties *complement* supervised pattern matching on regime detection. The geometric signal carries greatest value where supervised approaches lack historical precedent—precisely the novel crises where risk management matters most. Granger causality analysis (Section 6.8) provides partial evidence for temporal precedence of QFI Determinant over market volatility, though reverse causality dominates. Future work will pursue real-time hybrid deployment, cross-asset-class geometry (fixed income, commodities, FX), and learned operators via end-to-end optimization.

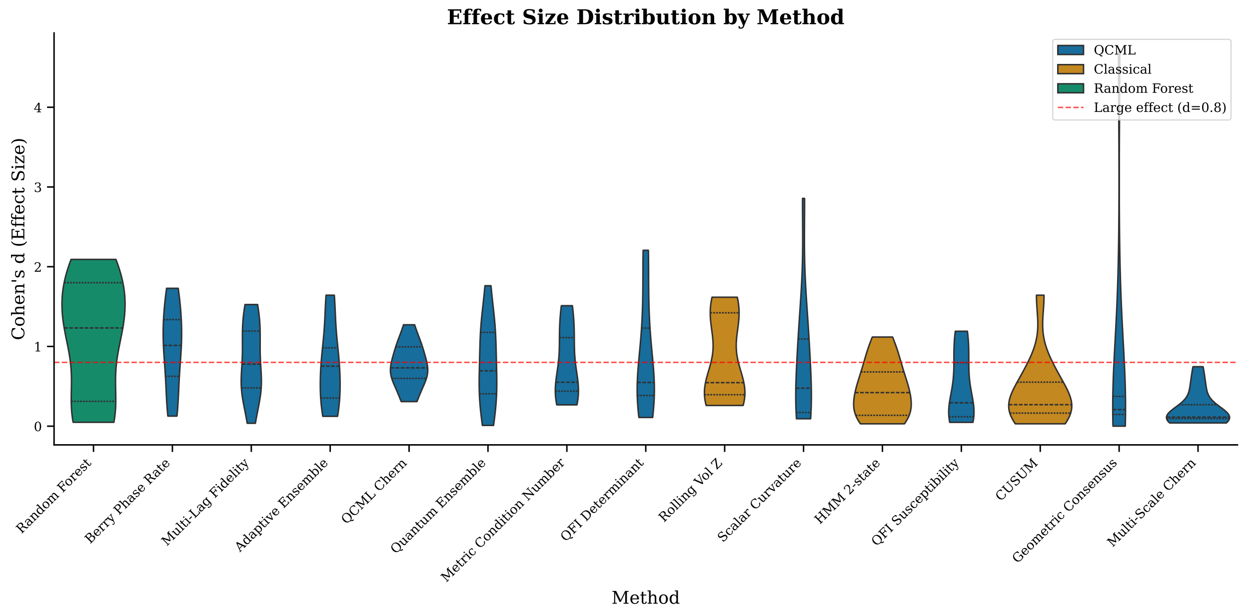

Figure 9: Violin plot of Cohen's d effect size distributions across 12 crises for all 16 methods. QCML methods shown in blue; classical baselines in gray. Horizontal dashed line at $d = 0.8$ marks the "large effect" threshold.

Data and Code Availability

Market data were obtained from the Polygon.io API under a paid subscription. The geometry library (`qcml_geometry/`) and all experiment scripts (`experiments/`) are available at `[repository URL upon acceptance]`. Crisis period definitions, hyperparameter configurations, and result JSON files are included for full reproducibility.

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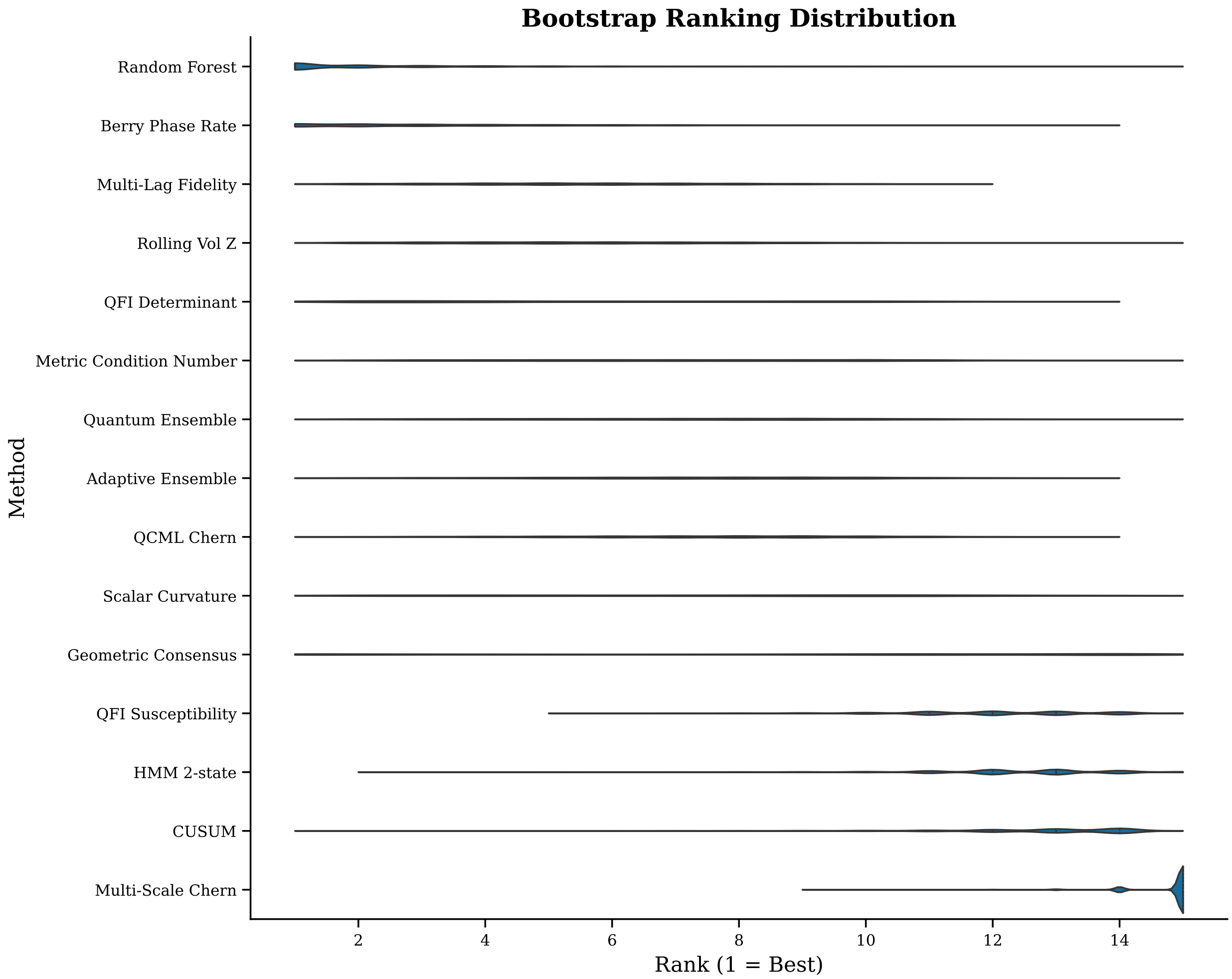

Figure 10: Bootstrap ranking distribution (n = 10,000) for all methods. Lower rank is better. Violin widths show posterior uncertainty.

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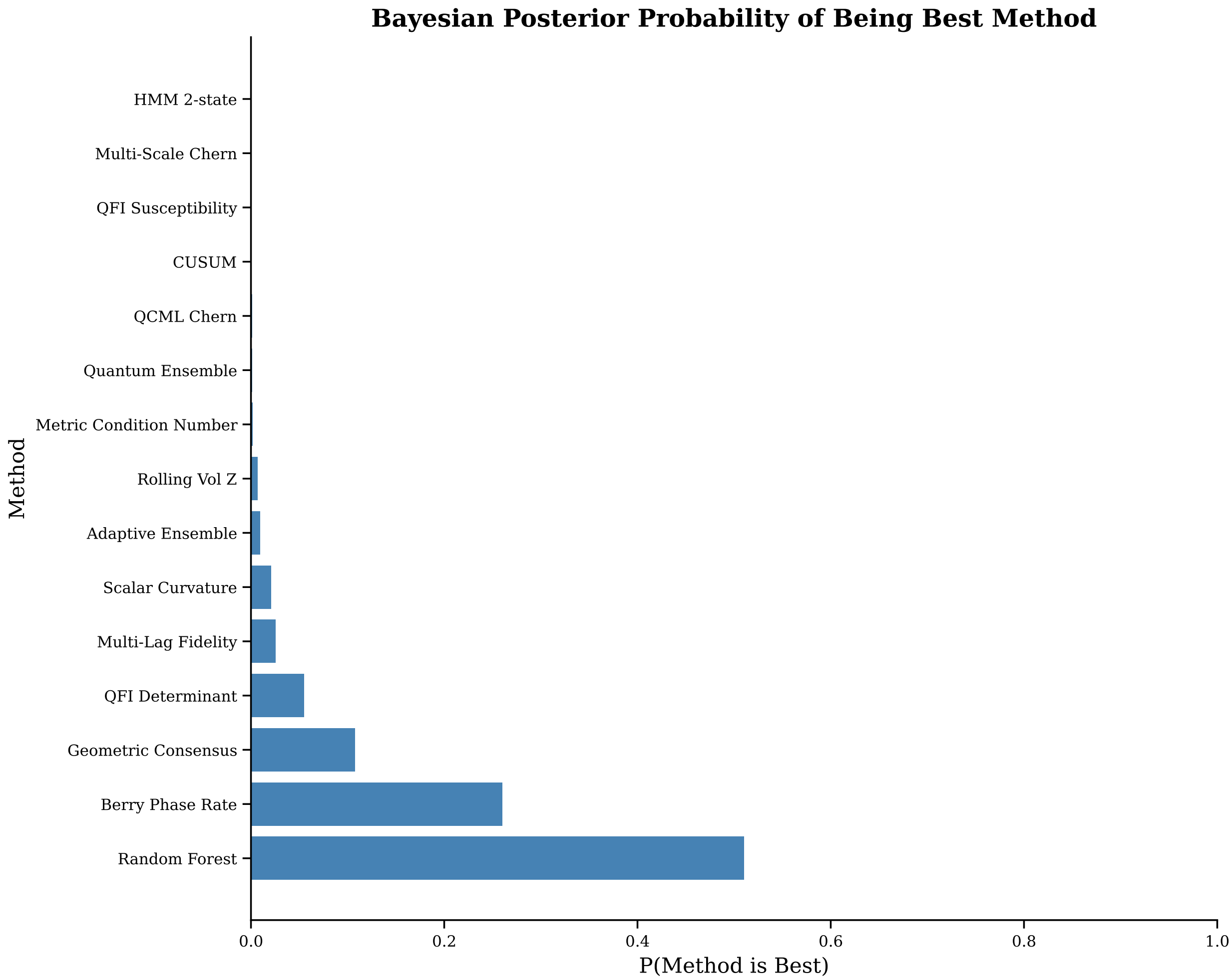

Figure 11: Bayesian posterior probability of each method being the best performer, estimated via bootstrap resampling of crises.

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Table 8: Default hyperparameters for the QCML regime detection pipeline.

Parameter	Description	Default
n	Hilbert space dimension	8
d	Number of PCA components	15
w	Chern rolling window	20 days
τ	Chern transition threshold	0.1
k	Indicator collapse threshold (σ)	2.0
W	Indicator rolling window	20 days
δ	Fidelity lag	1 day
Operator method	Hermitian operator construction	random
Normalization	Feature standardization	z-score + L^2
Multi-scale windows	Consensus Chern scales	$\{10, 20, 30, 50\}$
Scale weights	Consensus weights	$[0.15, 0.30, 0.30, 0.25]$
Persistence filter	Minimum consecutive spike days	2
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

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A Proofs of Formal Results

Proof of Theorem 2. Consider the eigenvalue equation $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$ subject to the normalization constraint $\langle\psi|\psi\rangle = 1$. Define the map $\Phi\colon \mathbb{R}^d \times \mathbb{C}^n \times \mathbb{R} \to \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = \big(H(x)\psi - E\psi,\ \|\psi\|^2 - 1\big)\,.$$

We have $\Phi(x_0, \psi_0, E_0) = 0$ at any point where $|\psi_0\rangle$ is the normalized ground state with eigenvalue E_0. The Jacobian of Φ with respect to (ψ, E) at (x_0, ψ_0, E_0) is

$$D_{(\psi,E)}\Phi = \begin{pmatrix} H(x_0) - E_0\, I & -\psi_0 \\ 2\psi_0^\dagger & 0 \end{pmatrix}.$$

The operator $H(x_0) - E_0\, I$ restricted to the orthogonal complement $\psi_0^\perp$ is invertible with smallest singular value $\Delta(x_0) > 0$ (the spectral gap). The full Jacobian is therefore invertible whenever $\Delta > 0$. By the implicit function theorem [28], $(x \mapsto \psi(x), x \mapsto E_0(x))$ are C^∞ functions of x on any open set where $\Delta > 0$.

Since $H(x)$ depends polynomially on x (it is a sum of squares of affine functions), Φ is C^∞ in all arguments. The regularity of g_{ab} and F_{ab} follows immediately from their definitions as derivatives of $|\psi(x)\rangle$, and all higher geometric quantities (Christoffel symbols, Ricci scalar, Chern numbers) are smooth as compositions of smooth functions. □

Proof of Theorem 3. The ground state $|\psi(x)\rangle$ defines a $U(1)$ principal bundle (or equivalently, a complex line bundle $\mathcal{L}$) over the parameter surface S. The Berry connection is the natural $U(1)$ connection on $\mathcal{L}$:

$$\mathcal{A}_a(x) = i\langle\psi(x)|\partial_a\psi(x)\rangle\,,$$

and the Berry curvature $F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a$ is its curvature 2-form.

By the Chern–Weil theorem [10, 29], the integral of the curvature 2-form over any closed oriented surface equals 2π times an integer:

$$\oint_S \frac{F}{2\pi} = c_1(\mathcal{L}) \in \mathbb{Z}\,,$$

where $c_1(\mathcal{L})$ is the first Chern class of the line bundle $\mathcal{L} \to S$. This integer classifies the topological type of $\mathcal{L}$: line bundles with different Chern numbers are topologically inequivalent and cannot be smoothly deformed into each other.

The invariance under smooth deformations follows from the fact that $c_1 \in H^2(S;\mathbb{Z})$ is a topological (cohomological) invariant: any smooth family of Hamiltonians $H_s(x)$ that maintains $\Delta > 0$ along S defines a smooth family of line bundles with the same Chern class. □

Proof of Theorem 4. Using first-order perturbation theory, the Berry connection can be written as:

$$\mathcal{A}_a = i\langle\psi_0|\partial_a\psi_0\rangle = i\sum_{n\neq 0}\frac{\langle\psi_n|\partial_a H|\psi_0\rangle}{E_0 - E_n}\langle\psi_0|\psi_n\rangle\,,$$

where $|\psi_n\rangle$ and E_n are the eigenstates and eigenvalues of $H(x)$. Taking the curl:

$$F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a = -2\,\mathrm{Im}\sum_{n\neq 0}\frac{\langle\psi_0|\partial_a H|\psi_n\rangle\langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}\,.$$

Taking absolute values and using the triangle inequality:

$$|F_{ab}| \leq 2 \sum_{n \neq 0} \frac{|\langle \psi_0 | \partial_a H | \psi_n \rangle| \, |\langle \psi_n | \partial_b H | \psi_0 \rangle|}{(E_n - E_0)^2} \, .$$

Since $E_n - E_0 \geq \Delta$ for all $n \neq 0$:

$$|F_{ab}| \leq \frac{1}{\Delta^2} \, 2 \sum_{n \neq 0} |\langle \psi_0 | \partial_a H | \psi_n \rangle| \, |\langle \psi_n | \partial_b H | \psi_0 \rangle| = \frac{K}{\Delta^2} \, ,$$

where K depends only on the matrix elements of $\partial_a H$ and $\partial_b H$ between eigenstates, not on the gap. This completes the proof. □

Proof of Proposition 5. Let g_{ab} have eigendecomposition $g = V \, \mathrm{diag}(\lambda_1, \ldots, \lambda_d) \, V^T$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_d$. The Riemannian volume element on the non-degenerate r-dimensional subspace is $\mathrm{vol}_r = \sqrt{\det(g|_r)}$, where $g|_r$ is the restriction of g to the span of the first r eigenvectors. Then

$$\det(g|_r) = \prod_{i=1}^{r} \lambda_i = \overset{+}{\det}(g) \, .$$

Thus $\det^+(g) = (\mathrm{vol}_r)^2$.

The interpretation follows directly:

1. When $\det^+(g) \to 0$, either the effective rank r decreases (eigenvalues crossing zero) or some eigenvalues approach zero, indicating dimensional collapse of the state space—the market state is confined to a lower-dimensional subspace, characteristic of phase boundaries where correlations increase.

2. Large $\det^+(g)$ indicates that the metric has full rank with large eigenvalues: the market state is sensitive to perturbations in all directions, characteristic of a stable, multi-factor regime where diverse risk factors are active. □

B Supporting Analyses

B.1 Ablation Study

Table 9 ranks all 10 geometry-induced observables as standalone detectors. The top three (Berry Phase Rate $d = 0.93$, QFI Determinant $d = 0.93$, Multi-Lag Fidelity $d = 0.84$) are used throughout the paper for fusion and multi-asset experiments. Ensemble methods (mean $d = 0.73$) perform comparably to the best single indicators, indicating that individual observables capture distinct aspects of regime transitions.

B.2 False Positive Rate Analysis

We evaluate specificity using four calm periods (2005–06, 2012–13, 2017H1, 2021H1) where VIX $<$ 15 and drawdown $<$ 5%. HMM achieves the highest ROC-AUC (0.975) due to its latent-state partition. Among QCML methods, Multi-Scale Chern (0.859) and QCML Chern (0.748) have the best specificity. Berry Phase Rate shows moderate specificity (0.461), reflecting a sensitivity–specificity trade-off: methods that fire early register as false positives under strict labeling.

Table 9: Individual geometric indicator contributions.

Indicator	Mean d	Rank
Berry Phase Rate	0.99	1
QFI Determinant	0.86	2
Multi-Lag Fidelity	0.84	3
Metric Condition No.	0.77	4
Geometric Ensemble	0.77	5
Scalar Curvature	0.75	6
QCML Chern	0.75	7
Geometric Consensus	0.66	8
QFI Susceptibility	0.49	9
Multi-Scale Chern	0.22	10

Table 10: False positive analysis: crisis vs. calm discrimination.

Method	ROC-AUC	PR-AUC	F1
HMM 2-state	0.975	0.735	0.928
Multi-Scale Chern	0.859	0.425	0.639
QCML Chern	0.748	0.276	0.595
Metric Cond. No.	0.745	0.313	0.538
Berry Phase Rate	0.461	0.189	0.341
QFI Determinant	0.300	0.138	0.339

B.3 Method Correlation

PCA of the 10 geometric score time series reveals that 8 components are needed for 90% variance (PC1 explains 32.1%, first three: 57.3%), indicating high-dimensional structure. Berry Phase Rate and QFI Determinant are negatively correlated (Spearman $\rho = -0.25$), confirming they capture distinct geometric signals and supporting their joint use in fusion architectures.